

Solution Requirements

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user authentication is required. The application is publicly accessible for HDI prediction.	Low
2	Authorization Levels	All users have equal access to enter inputs and generate predictions.	Low
3	External Interfaces	Accepts user input through a Flask web interface and uses the trained Machine Learning model for prediction.	High
4	Transaction Processing	Receives Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita, processes the data, and predicts the HDI score.	High
5	Reporting	Displays the predicted HDI score and corresponding development category (Low, Medium, High, or Very High).	High
6	Business Rules	Prediction is generated only when all required input values are provided. The HDI category is assigned based on the predicted score.	High
7	Compliance to Laws or Regulations	The system does not collect or store personal information and follows basic data privacy practices.	Medium
8	Other	Provides a simple, user-friendly interface with quick and accurate predictions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate HDI predictions quickly.	Prediction generated within 2 seconds.
2	Scalability	The application should support multiple users without affecting prediction accuracy.	Supports multiple concurrent users.
3	Security & Data Privacy	User input is processed securely and no personal data is stored.	No sensitive data stored; secure input handling.
4	Reliability & Availability	The application should consistently provide accurate predictions whenever accessed.	99% availability during normal operation.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on desktop and mobile devices.	User-friendly interface with responsive design.
6	Other	The application should be easy to maintain and update with new datasets or improved machine learning models.	Modular Python code with maintainable architecture.